Nomenclature

Coupling variables and numerical parameters

Symbol	Name and physical meaning	Unit	Constraint or adopted value
$q_{prd}^{(k)}$	Prescribed interface motion vector applied to Abaqus in coupling trial k; contains three translations and three rotations.	m; rad	$q_{prd}^{(0)}$ is the initial prediction; Updated by IQN-ILS/fixed under-relaxation
$q_{OF}^{(k)}$	Interface motion returned by the non-committing OpenFAST trial.	m; rad	Approaches $q_{prd}^{(k)}$ at convergence.
$\boldsymbol{u},\boldsymbol{\theta}$	Interface translation vector and rotation vector; subscripts prd and OF identify prescribed and OpenFAST motions.	m; rad	Translations and rotations are expressed in meters and radians, respectively.
$F_{ABA}^{(k)}$ $F_{OF}^{(k)}$	Generalized foundation reactions evaluated by Abaqus and represented in OpenFAST, respectively.	N; N·m	Three force components and three moment components; the two reaction vectors coincide as the interface residual approaches zero.
$\boldsymbol{f}_{ABA}^{(k)}$ $\boldsymbol{m}_{ABA}^{(k)}$	Abaqus reaction force vector and reaction moment vector.	N; N·m	Reactions from Abaqus under the adopted interface motions.
$\Delta R_{NL}^{(k)}$	Nonlinear load correction after subtracting the reference linear contribution.	N; N·m	$\Delta R_{NL}^{(k)}=F_{ABA}^{(k)}-K_0 q_{prd}^{(k)}$
K_0	Reference linear foundation stiffness matrix retained in SubDyn.	N/m; N/rad; N·m/m; N·m/rad	The OC6 verification cases use the prescribed benchmark stiffness; LF-spring, MF-1D, and HF-MP use the initial linearized HF-MP stiffness; HF-PBH uses its independently determined initial stiffness.
$\boldsymbol{r}^{(k)}$	Unscaled interface motion residual.	m; rad	Contains both translational and rotational residuals.
$\tilde{\boldsymbol{r}}^{(k)}$	Equivalent interface displacement residual obtained by scaling the rotational components of the unscaled residual.	m	Convergence requires its Euclidean norm to be less than ε_{abs}.
$\boldsymbol{S}$	Diagonal scaling matrix converting rotational residuals into equivalent translational displacements.	Mixed	$S=diag(1, 1, 1, L_c, L_c, L_c)$.
L_c	Characteristic length used to scale rotational interface residuals.	m	$L_c=D/2$, where D is the representative foundation diameter adopted for residual scaling.
ε_{abs}	Absolute tolerance for the scaled interface residual norm.	m	10^{-7} m in this study.
$\boldsymbol{V}^{(k)}$, $\boldsymbol{W}^{(k)}$	Secant-history matrices containing differences in interface residuals and corresponding OpenFAST interface motions, respectively.	m; rad	Secant information from the current physical time step and four preceding time steps is retained.

Symbol	Name and physical meaning	Unit	Constraint or adopted value
$\boldsymbol{c}^{(k)}$	Least-squares coefficient vector used in the IQN-ILS update.	1	Unconstrained coefficients obtained from the current least-squares problem.
R_{ii}	Diagonal entry of the upper-triangular matrix obtained from QR decomposition of the secant-history matrix.	Mixed	The adopted numerical rejection criterion is $\lvert R_{ii}\rvert<10^{-10}$.
ω	Fixed under-relaxation factor used when valid IQN-ILS secant information is unavailable.	1	ω = 0.02 in this study. Distinct from the constitutive parameter ω = 0.1 in Appendix C.1.
t_n, t_{n+1}	Committed physical time and target time of the next coupling step.	s	Coupling interval is 0.01 s in this study.
t_a, t_r	Latest accepted Abaqus internal time and current internal evaluation time.	s	$t_n \le t_a \le t_r \le t_{n+1}$; interpolation requires $t_a < t_{n+1}$.
q_n q_a $q_{t_r}^{(k)}$	Interface motions at t_n, t_a and t_r, respectively.	m; rad	Linear interpolation over the remaining Abaqus interval; $q_a = q_n$ when $t_a = t_n$.

Geometry, loading and soil condition

Symbol	Name and physical meaning	Unit	Constraint or adopted value
D	Monopile outer diameter; representative foundation diameter when used in L_c.	m	Monopile diameters are 9 m for OC6 and 6 m for the NREL 5 MW case study.
L	Embedded monopile length.	m	45 m for OC6 and 60 m for the case study.
z	Depth coordinate measured vertically below the mudline.	m	$z = 0$ at the mudline, positive downwards.
Z_R	Depth of the prescribed monopile rotation center in MF-1D.	m	$Z_R = 0.8L = 48$ m in the case study.
H	Length of the embedded pile below the MF-1D rotation point.	m	$H = L - Z_R$; $H = 12$ m and $H/D = 2$ in the case study.
V_{hub}	Mean wind speed at hub height defining the loading condition.	m/s	9.06 m/s for the prescribed OC6 wind-loading histories; 12 m/s for the case study.
H_s	Significant wave height.	m	1.25 m for OC6 and 1.7 m for the case study.
T_p	Wave spectral peak period.	s	5.5 s for OC6 and 5.88 s for the case study.
F_u	Reference ultimate lateral load of the monopile used to normalize the centrifuge cyclic loading.	N	One-way cyclic loading at $0.25F_u$ for 100 cycles, applied 8 m above the mudline in the reference centrifuge tests.
s_u	Undrained shear strength of the soil.	kPa	$s_u(z) = (1.54\ \text{kPa/m})z$ for the normally consolidated Malaysian kaolin in the case study.
s_{u0}	Undrained shear strength at the MF-1D rotation point.	kPa	$s_{u0} = s_u(Z_R) = 73.92$ kPa, with $Z_R = 48$ m.
k	Gradient of undrained shear strength with depth.	kPa/m	1.54 kPa/m for the adopted increasing profile; Distinct from the coupling-iteration index k.

Soil reaction model

Symbol	Name and physical meaning	Unit	Constraint or adopted value
τ	Mobilized shear stress in the adopted NGI-ADP element stress‑strain response.	kPa	For the positive backbone in Eq. (B.1), $0 \leq \tau/s_u \leq 1$, with full mobilization at $\gamma^p = \gamma_f^p$.
$\gamma, \gamma^e, \gamma^p$	Total, elastic and plastic shear strains, respectively.	1	$\gamma = \gamma^e + \gamma^p$ and $\gamma^e = \tau/G_{max}$
γ_f^p	Plastic shear strain at full mobilization of the undrained shear strength.	1	$\gamma_f^p = 0.15$ for Malaysian kaolin.
G_{max}	Small-strain shear modulus used for the elastic strain component.	kPa	$G_{max}/s_u = 500$ for Malaysian kaolin.
G_{max}/s_u	Small-strain shear modulus to undrained strength ratio.	1	$G_{max}/s_u = 500$ for Malaysian kaolin.
p, p_u	Mobilized and ultimate lateral soil resistance per unit pile length, respectively.	N/m	$p = (\tau/s_u)p_u$ and $p_u = N_p s_u D$.
y	Local lateral pile displacement associated with a p-y spring.	m	$y = D(\xi_y^e \gamma^e + \xi_y^p \gamma^p)$ for the adopted soil-reaction backbone.
α	Pile-soil interface roughness factor in the MF-1D reaction model.	1	$0 \leq \alpha \leq 1$, ranging from smooth to fully rough interaction; $\alpha = 1$ in the case study; This parameter is distinct from the Coulomb friction coefficient used in the 3D FE models.
N_p $N_{p,w}$ $N_{p,f}$	Adopted, shallow-wedge and full-flow lateral bearing-capacity factors.	1	$N_p = min(N_{p,w}, N_{p,f})$; $N_{p,f} = 9.14 + 2.8\alpha = 11.94$ in the case study.
ξ_y^e, ξ_y^p	Elastic and plastic scaling factors mapping element shear strain to normalized lateral displacement.	1	Use the coefficients associated with the adopted soil failure mechanism; These coefficients depend on depth and roughness.
$\xi_{y,w}^e$ $\xi_{y,w}^p$	Elastic and plastic displacement scaling factors for the shallow-wedge mechanism.	1	Depth-dependent empirical relations in z/D; Applicable within the shallow-wedge formulation.
$\xi_{y,f}^e$ $\xi_{y,f}^p$	Elastic and plastic displacement scaling factors for the full-flow mechanism.	1	$\xi_{y,f}^e = 2.8$ and $\xi_{y,f}^p = 1.35 + 0.25\alpha = 1.60$ in the case study.
M_R, $M_{R,ult}$	Mobilized and ultimate resisting moments condensed at the monopile rotation point, respectively.	N·m	Ultimate resistance depends on geometry and soil strength.
θ_R	Pile rotation associated with the concentrated moment spring.	rad	$\theta_R = \xi_\theta^e \gamma^e + \xi_\theta^p \gamma^p$.
ξ_θ^e, ξ_θ^p	Elastic and plastic scaling factors mapping element shear strain to pile rotation.	1	Functions of H/D; 1.27 and 0.72, respectively, in the case study.
ψ	Geometric angle in the ultimate moment-resistance expression.	rad	$\psi = arcsin[D/\sqrt{D^2 + 4H^2}]$

Constitutive and finite element parameters

Symbol	Name and physical meaning	Unit	Constraint or adopted value
e, e_0	Current and initial soil void ratios.	1	The initial profile depends on the effective stress state.
p'	Mean effective stress.	kPa	Positive in compression.
λ	Slope of the normal compression line.	1	MCC compression parameter; λ = 0.26 for Malaysian kaolin.
κ, κ_0	Current and initial unloading or swelling slopes, respectively.	1	κ_0 = 0.015 for Malaysian kaolin.
N	Intercept of the normal compression line.	1	N = 2.73 for Malaysian kaolin.
M_t	Critical-state stress ratio of the clay constitutive model.	1	M_t = 1.1 for Malaysian kaolin.
v	Poisson's ratio.	1	Soil, pile and bucket: 0.3; grout: 0.2.
C	Material constant controlling nonlinear elastic behavior.	1	C = 10 for Malaysian kaolin.
ω	Material constant controlling nonlinear elastic behavior.	1	ω = 0.1 for Malaysian kaolin. Distinct from the fixed coupling under-relaxation factor.
X, X_s	Dimensionless stress-history variables relating current and stress-reversal states.	1	Defined by the adopted nonlinear elastic formulation.
K_p, $\overline{K}_p$	Plastic moduli at the current stress point and its corresponding image stress point on the bounding surface, respectively.	kPa^3	State-dependent constitutive quantities defined using the adopted yield function.
$\overline{p_0'}$	Preconsolidation pressure defining the size of the bounding surface.	kPa	State-dependent hardening variable governing the size of the bounding surface.
p_a	Atmospheric reference pressure in the plastic-modulus relation.	kPa	p_a=101 kPa in the adopted model.
R	Radial mapping ratio between the current and image stress points	1	State-dependent ratio.
n, n_0	Current and initial exponents of the radial-mapping rule.	1	$n=n_0 exp(\zeta \varepsilon_q^p/\varepsilon_r)$; n_0=50 for Malaysian kaolin.
ζ	Parameter controlling the strain dependence of the mapping exponent.	1	ζ=−1 for Malaysian kaolin.
ε_q^p	Accumulated deviatoric plastic strain governing cyclic constitutive evolution.	1	Non-negative accumulated internal variable; Depends on the preceding loading history.
ε_r	Reference strain normalizing the accumulated deviatoric plastic strain.	1	ε_r>0; 0.01 for Malaysian kaolin.
k_0	Initial permeability coefficient.	m/s	Related to e_0 through the calibrated logarithmic relation.

Symbol	Name and physical meaning	Unit	Constraint or adopted value
E	Young's modulus of the structural materials.	GPa	Pile: 223 GPa; Bucket: 210×10^3 GPa; Grout: 50.7 GPa; The bucket modulus is artificially increased to approximate rigid-bucket behaviour.
EI	Pile flexural rigidity.	$N\cdot m^2$	The pile modulus is adjusted to reproduce the prototype flexural rigidity reported by Lai et al. (2020).
μ	Coulomb friction coefficient governing tangential soil-foundation contact in the high-fidelity FE models.	1	$\mu = 0.3$ for HF-MP and HF-PBH; Distinct from the interface roughness factor.

Response quantities and stiffness identification

Symbol	Name and physical meaning	Unit	Constraint or adopted value
u_x, θ_y	Fore-aft displacement and rotation about the global y-axis, respectively.	m; rad	-
Δq_i, ΔF_i	Mean-centered two-component motion and reaction vectors for sample i within a moving window.	m; rad / N; N·m	-
$\bar{u}_x$ $\bar{\theta}_y$ $\bar{f}_{\mathrm{ABA},x}$ $\bar{m}_{\mathrm{ABA},y}$	Window-mean lateral displacement, rotation, lateral reaction force, and reaction moment, respectively.	m; rad; N; N·m	Evaluated within each stiffness-identification window.
K_{uu}	Operational lateral foundation stiffness identified by a symmetric local fit.	N/m	Window-dependent effective stiffness.
$K_{u\theta}$	Operational sway-rocking cross-coupling stiffness.	N/rad	Symmetry is imposed in the fitted 2 × 2 stiffness matrix.
$K_{\theta\theta}$	Operational rocking foundation stiffness.	N·m/rad	Window-dependent effective stiffness.
$R_{s/d}$	Ratio of shallow to deep resisting-moment contributions.	1	Values above or below 1 indicate greater shallow or deep contribution under the adopted resistance convention.